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Project Submission: Developing a Centralized Data Lake Solution Using Historical Project Data on the Construction Check Cloud-Based Platform

1. Organization Background & Information

About Construction Check

Construction Check is a digital services platform that draws on the expertise and experience of seasoned cost estimating professionals. Construction Check was founded to address critical shared pain in the professional construction cost estimating industry brought on by a dwindling pool of skilled workers, increased certification demand, and a lack of infrastructure to expedite the acquisition of objective and unbiased construction cost estimates.

About Project Cost Solutions (PCS)

Project Cost Solutions (PCS) is a leader in construction cost estimating, providing owners with accurate project cost insights derived from years of historical data across various project types. PCS specializes in cost planning, budgeting, and financial forecasting, helping clients make data-driven decisions that enhance project efficiency and cost management.

2. Problem Statement & Objective

Problem Statement:

- Fragmented and inconsistent historical cost data makes it difficult to analyze trends and extract meaningful insights.
- A lack of centralized, standardized cost data prevents stakeholders from making informed, data-driven decisions.
- Cost estimators, owners, developers, architects, engineers, and government agencies need a shared platform for real-time cost benchmarking and forecasting.
- Traditional cost estimation methods struggle with early-stage (Class 5) estimates, leading to significant budget uncertainty.

Objective:

The goal of this project is to develop a collaborative, cloud-based data lake leveraging PCS's historical cost data as the foundation. The Construction Check platform will serve as an industry-wide echo system, where architects, engineers, developers, government agencies, and owners can contribute their own project and cost estimate data while accessing machine learning-driven cost predictions for benchmarking and decision-making.

By centralizing cost data across sectors, Construction Check will enhance parametric cost estimation by enabling AI-driven cost predictions that improve early-stage estimates. The platform will also provide an adaptive UX/UI interface that supports **Class 5 cost estimation** (with an accuracy range of -50% to +100%).

3. Desired/Required Skills

Data Analysts:

- **Data Analysis & Modeling:** Ability to structure and analyze historical project data to identify trends and improve cost estimation accuracy.
- **SQL or Similar Querying Languages:** Proficiency in extracting and manipulating data from the Construction Check platform's database.
- **Machine Learning for Cost Estimation:** Understanding of regression models, random forests, and neural networks to predict cost parameters.

Software Engineers:

- **Programming Languages:** Experience with web development technologies compatible with Microsoft Azure (e.g., C#, Python).
- **API Integration:** Ability to connect the data lake with Construction Check's existing cost estimation tools and reporting functionalities.
- **User Interface (UI) Design:** Capable of developing an intuitive UI for users to interact with cost data and visualization dashboards.
- **Data Security:** Experience implementing robust security measures to ensure compliance and protect user data.

4. Names of Key Contacts

- **Leanda Thomas** | Phone: 404-566-7502 | Email: leanda.thomas@constructioncheck.io
- **Gina Bennett** | Phone: 770-265-4005 | Email: gina.bennett@constructioncheck.io

5. Nature, Size, and Availability of Data Set to Be Analyzed

The project will analyze historical project cost data from PCS as the initial foundation, with the goal of expanding to a shared cost database contributed to by industry professionals across various sectors.

Data Sources:

- **PCS Historical Data:** Established cost estimation benchmarks across multiple industries.
- **User-Contributed Data:** Cost estimates and project details from architects, engineers, developers, government agencies, and owners.
- **Project Data:**
 - Project type (Infrastructure, Commercial, Government, Private Development)
 - Project cost breakdowns (Labor, Material, Equipment)
 - Historical cost estimates vs. actual cost outcomes
 - Benchmarking data for cost trends across different regions and building types

Data Availability Considerations:

- Cost data exists in structured and semi-structured formats (Excel, SQL databases, PDFs).
- The data will be ingested into an Azure-based environment for processing and analysis.
- Data privacy and compliance measures will be implemented to ensure secure access.

6. Technologies/Tools to Be Used

Since Construction Check operates on Microsoft Azure, the following Azure services and data tools will be leveraged:

- **Azure Synapse Analytics:** Centralized storage for structured and semi-structured cost data.
- **Azure Data Factory:** Manages ETL processes for data ingestion.
- **Azure Databricks:** Transforms and standardizes cost data to ensure consistency.
- **Power BI:** Provides real-time dashboards and data visualization tools.
- **Azure AD:** Implements role-based access control to maintain data security.
- **SQL-based Databases:** Stores historical cost data and supports advanced queries.
- **Azure Machine Learning:** Enables deployment of machine learning models for parametric cost estimation.

7. UX/UI Interface for AI-Powered Cost Estimation

Once the centralized data lake is operational, Construction Check users will have access to an AI-powered generative interface for cost estimation. The system will include the following key features:

Engineering Prompt for LLM Implementation

System Role:

- A highly specialized **Generative AI model** designed to assist both technical and non-technical users in estimating project costs.
- The model's primary objectives are to classify project types, generate adaptive questions based on user expertise, and provide **Class 5 cost estimates** with an accuracy range of -50% to +100%.

User Context and Input

1. User Classification:

- Ask the user: "Are you a technical or non-technical user?" Tailor all subsequent interactions accordingly:
 - **Technical Users:** Use precise and industry-relevant terminology.
 - **Non-Technical Users:** Simplify language and guide with accessible terms.

8. Expected Benefits & Impact

For Construction Check and Industry Stakeholders:

- **Industry-Wide Data Collaboration:** Creates a shared ecosystem where users contribute cost data for enhanced industry benchmarking.
- **Improved Cost Estimation Accuracy:** Standardized data improves cost predictability and financial planning.
- **Operational Efficiency:** Streamlined cost data processing reduces redundancies.
- **Scalability for Industry Use:** The system can be expanded to benefit multiple stakeholders beyond PCS.
- **Class 5 Cost Estimation Accuracy:** Leveraging parametric cost estimating models enhances decision-making for early-stage projects.

For Students Participating in the Project:

- **Real-World Experience:** Work on a high-impact data analytics project applied in the construction industry.

- **Practical Application of Skills:** Apply knowledge in SQL, data modeling, machine learning, and cloud computing.
- **Collaboration with Industry Professionals:** Gain exposure to cost estimation methodologies and data-driven decision-making.
- **Portfolio Development:** Contribute to a real-world system implementation that can be showcased in job applications.

9. Next Steps

1. Conduct an internal audit of existing PCS cost data to assess completeness and usability.
2. Develop a structured database schema for ingestion into Azure Synapse Analytics.
3. Implement ETL processes using Azure Data Factory and validate initial datasets.
4. Train machine learning models for parametric cost estimation and validate accuracy.
5. Deploy solution on Construction Check's cloud platform for real-world testing.
6. Gather feedback and refine data standardization methodologies.
7. Expand data integration capabilities to include architects, engineers, developers, government agencies, and owners as active contributors.

This project represents a significant advancement in collaborative, machine learning-driven parametric cost estimation, creating an industry-wide echo system that enables shared data contribution and AI-powered cost intelligence for all stakeholders.